Indistinguishable at the Top: A Statistical Audit of the BLP-2025 Bangla Hate Speech Shared Task

Gazi Shoaib
Jahangirnagar University
Savar, Dhaka, Bangladesh
gazishoaib953@gmail.com

Abstract

Shared-task leaderboards routinely report scores to four decimal places and rank systems accordingly, but rarely report whether those ranks are distinguishable from noise. We audit the BLP-2025 Bangla Hate Speech Identification shared task, whose data, gold test labels, and evaluation scripts are fully public. We find that in all three subtasks the top five systems are statistically indistinguishable: the top-five spread is 0.34–0.80 percentage points against a 95% confidence interval half-width of 0.85–0.86 points, and for the leader-versus-runner-up gap to reach $p < .05$ under McNemar's test the two systems would have to disagree on under 1.2% of test items. We then test, and reject, the hypothesis that annotation noise caps performance: three-way majority voting over annotations with $\kappa = 0.71$ yields gold labels we estimate to be ≈97% reliable, leaving a 24-point gap to the best system. The binding problem is instead the metric. Because SEXISM comprises 29 of 10,200 test items, a system that detects every sexist comment and one that detects none differ by 0.28 points of micro-F1—below the 0.44-point sampling noise floor. Training ten systems to obtain real predictions, we measure pairwise discordance of 3.6–36.3% and show that the minimum detectable gap is 0.37 pp even in the best case—above every reported top-two gap—while a pair separated by 0.02 pp wins only 51.5% of bootstrap resamples. No system in our pool detects more than 3 of the 29 sexist comments, and five detect none. All analysis code is released.

1 Introduction

A shared task converts a research question into a ranking. The ranking is what participants optimise, what system-description papers narrate, and what subsequent work cites as the state of the art. Yet the NLP community has known for some time that small score differences on a fixed test set are frequently indistinguishable from sampling noise (Reimers and Gurevych, 2017; Dror et al., 2018; Card et al., 2020), and that benchmarks themselves carry label errors that destabilise the orderings they induce (Northcutt et al., 2021; Bowman and Dahl, 2021).

Jin et al. (2026) recently applied this scrutiny to text-to-SQL, manually re-examining two widely used benchmarks and reporting annotation error rates of 52.8% and 66.1%. Their central observation is not that any single benchmark is bad but that *nobody had checked*. We ask the same question of a low-resource benchmark, where the stakes are arguably higher: resources are scarcer, so a single shared task shapes a larger fraction of subsequent work.

We audit BLP-2025 Task 1, Bangla Hate Speech Identification (Hasan et al., 2025), a well-run shared task that attracted over 160 participants and—critically for our purposes—released its data, its gold test labels, and its scorer. Our contributions:

1. In all three subtasks the reported top five are statistically indistinguishable (§4).

2. We estimate the annotation ceiling from the reported inter-annotator agreement and *reject* the hypothesis that label noise explains the observed 0.736 plateau (§6).

3. We show that micro-F1 on this label distribution is formally incapable of registering performance on the SEXISM class (§7).

4. We identify 342 test items whose labels are internally inconsistent (§8).

We stress that the dataset survives our integrity checks well (§8), and that this audit was only possible *because* the organisers released everything needed to run it.

[1] Code and analysis scripts: github.com/gazishoaib33/blp25-audit

2 Related Work

Significance in NLP. Reimers and Gurevych (2017) showed that single-run score comparisons are unreliable; Dror et al. (2018) surveyed appropriate tests; Card et al. (2020) demonstrated that many NLP experiments are badly underpowered for the effect sizes they claim. Our §4 applies this lens to a shared-task leaderboard rather than to a single paper's ablations.

Benchmark quality. Northcutt et al. (2021) found pervasive label errors across ten popular test sets and showed that correcting them reorders model rankings; Jin et al. (2026) performed the analogous audit for text-to-SQL. Vidgen and Derczynski (2020) survey quality problems specific to abusive-language corpora, and Sap et al. (2019) show how annotation artifacts translate into deployed bias. We inherit the methodology of Jin et al. (2026) while closing a gap in it: that work reports neither an annotator count, an agreement statistic, nor an adjudication protocol.

Bangla NLP. Resources for Bangla have expanded rapidly (Bhattacharjee et al., 2022), and the BLP workshop series has become the field's principal venue (Hasan et al., 2025; Raihan et al., 2025). To our knowledge no Bangla benchmark has previously been subjected to a quantitative audit.

3 The Shared Task

BLP-2025 Task 1 (Hasan et al., 2025) released BanglaMultiHate, 50,746 YouTube comments from a Bangla news channel spanning 19 topics, annotated by 35 native speakers with three independent labels per item resolved by majority vote. Reported Fleiss' κ (Fleiss, 1971) is 0.71 for hate type, 0.79 for target, and 0.84 for severity. Splits are 35,522 train / 2,512 dev / 2,512 dev-test / 10,200 test. Three subtasks are evaluated by micro-F1: 1A predicts hate type (6 classes), 1B target (5 classes), and 1C all three dimensions jointly.

A reproducibility hazard. The literal string `None` is the majority class (56.4% of 1A), and `pandas.read_csv` silently coerces it to `NaN` by default—a trap we hit ourselves. Loads must pass `keep_default_na=False`; the majority baseline should then reproduce the organisers' reported 0.5638, which ours does exactly.

	Leader vs. 2nd		Max. discordance
Subtask	gap (pp)	items	for $p < .05$
1A	0.17	17	78 (0.77%)
1B	0.21	21	119 (1.17%)
1C	0.14	14	53 (0.52%)

Table 1: Two systems would have to agree on more than 98.8% of the 10,200 test items for the observed leader-versus-runner-up gap to be significant.

4 Are the Rankings Meaningful?

Confidence intervals. Figure 1 plots the reported scores with 95% Wilson intervals (Wilson, 1927). The interval half-width is 0.85–0.86 points, while the entire top-five spread is 0.34 points (1A), 0.39 (1B), and 0.80 (1C). In all three subtasks, five of five teams have intervals overlapping the leader's lower bound.

Wilson intervals treat the two scores as independent, which understates precision for systems evaluated on the *same* items. We therefore turn to the paired test.

McNemar sensitivity. For paired predictions, McNemar's test (McNemar, 1947) depends only on the discordant counts b and c: the score gap is $d = (b-c)/n$, and significance at $p < .05$ requires $|b-c| > 1.96\sqrt{b+c}$. Solving for the largest discordance $b+c$ compatible with significance at the observed gap gives Table 1.

How plausible is such agreement? Rather than assume, we measure it (§5).

5 How Much Do Systems Actually Disagree?

The bound in §4 is only as good as its assumption about discordance, so we measure it. We trained ten systems on the official train split and evaluated each once on the official test split, varying feature space, classifier family, regularisation, and class weighting—the axes along which real submissions differ. Micro-F1 ranges 0.4984–0.6776; these are classical models and do not reach the 0.7362 of the shared task's transformer ensembles (see Limitations).

Measured discordance. Across all 45 pairs, the two systems make different predictions on a median of 32.0% of items (range 4.4–48.5%), and McNemar's discordant cells $b+c$ cover a median of 25.5% (range 3.6–36.3%). Restricting to the ten pairs within 1 pp of each other—the regime the

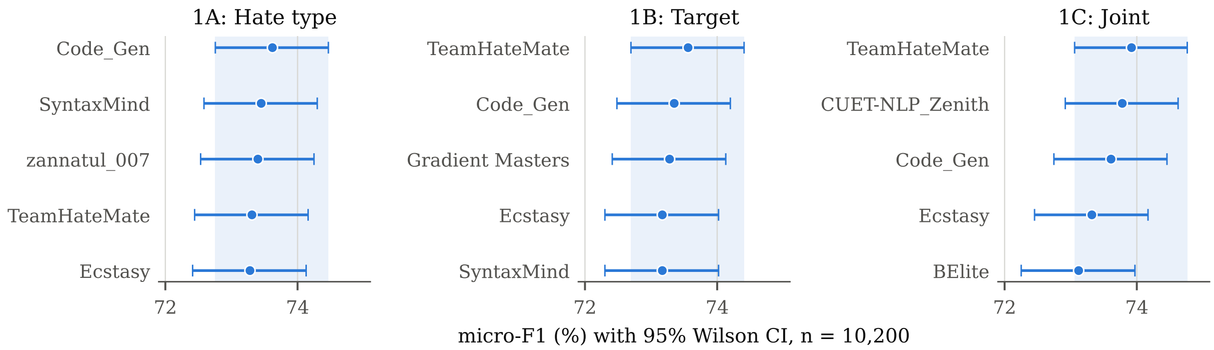

Figure 1: Reported micro-F1 for the top five teams in each subtask, with 95% Wilson confidence intervals at $n = 10{,}200$. In every subtask, all five teams fall within the leader's own interval.

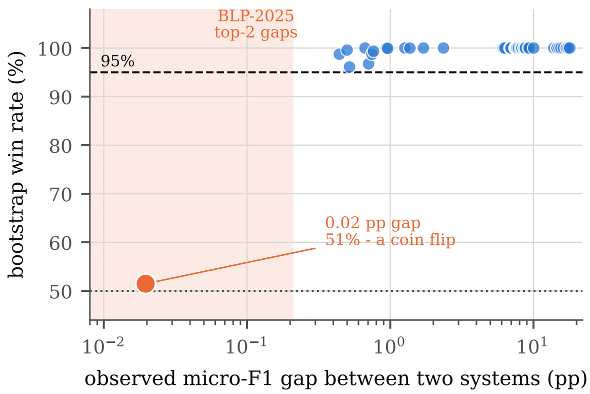

Figure 2: Bootstrap win rate against observed micro-F1 gap for all 45 pairs of our ten systems. The only pair whose gap falls in the BLP-2025 range is a coin flip (51.5%). Every pair we could resolve reliably had a gap of at least 0.44 pp.

BLP leaderboard occupies—median discordance is still 8.8%, or 895 items. **The lowest discordance we observed anywhere was 3.6%, against the <0.8% that Table 1 shows would be required.**

Minimum detectable difference. Inverting McNemar at each observed discordance level gives the smallest gap that could be resolved (Table 2). Even at our lowest observed discordance the minimum detectable gap is 0.37 pp; all three BLP top-two gaps (0.17, 0.21, 0.14 pp) fall below it.

Discordance level	% of n	Min. gap
lowest observed	3.6	0.37 pp
median, near-tied pairs	8.8	0.57 pp
median, all pairs	25.5	0.98 pp
highest observed	36.3	1.17 pp
BLP-2025 top-two gaps		0.14–0.21 pp

Table 2: Smallest micro-F1 gap resolvable at $p < .05$, given observed discordance. Every BLP top-two gap is below every threshold.

A necessary qualification. Small gaps are not automatically meaningless: our own top two differ by 0.44 pp with only 4.1% discordance and hold their order in 98.7% of 4,000 resamples. The defensible claim is conditional—a gap is uninformative when smaller than $1.96\sqrt{b+c}/n$—so what condemns the BLP gaps is their size *relative to plausible discordance*, not their size alone. Figure 2 shows the transition: the one pair inside the BLP range wins 51.5% of resamples, while every pair we could resolve had a gap of at least 0.44 pp.

Micro-F1 hides the systems that behave best. Five of our ten systems score exactly 0.000 F1 on SEXISM, detecting none of its 29 instances; the best detects 3, for an F1 of 0.115. (Only one system never emits the label at all; the others emit it but never correctly.) That best-performing system on SEXISM also has the best macro-F1, 0.5019—and ranks only *fourth* of ten on micro-F1. Micro and macro rankings agree weakly (Kendall $\tau = 0.467$, $p = .073$). Optimising the reported metric does not merely fail to reward rare-class detection: in our pool it actively demotes the system that does it best.

6 Is Annotation Noise the Ceiling?

A natural explanation for a plateau at 0.736 is that the labels themselves are only that reliable. We test this and find it false.

From the reported κ and the observed class marginals ($P_e = 0.3887$), pairwise annotator agreement is $P_o = \kappa(1 - P_e) + P_e = 0.823$. Modelling each annotator as correct with probability α and otherwise mistaken, we solve for α under two error models: errors scattered uniformly across the remaining classes, and errors maximally correlated onto a single confusable class—the pessimistic

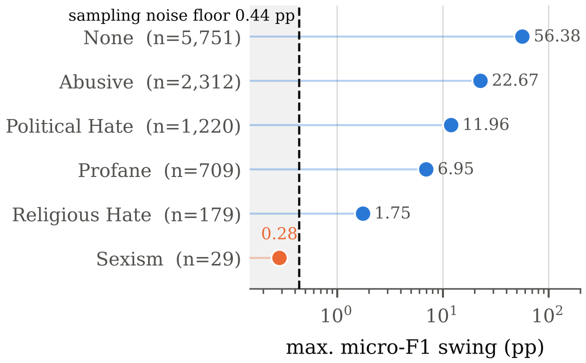

Figure 3: The largest micro-F1 change any system can obtain by going from zero to perfect recall on each class. Sexism falls below the sampling noise floor: the metric cannot detect whether a system handles it at all.

case, since correlated errors defeat majority voting. We then compute the probability that the majority of three annotators recovers the true label.

Dimension	κ	P_o	α	Gold acc.
Type	0.71	0.823	.906 / .902	**.975 / .973**
Target	0.79	0.872	.933 / .931	.987 / .986
Severity	0.84	0.902	.950 / .948	.993 / .992

Table 3: Estimated per-annotator accuracy α and majority-vote gold accuracy, under uniform / correlated error models.

Both models place the ceiling near 0.97 (Table 3). Three-way majority voting is strikingly effective at cleaning $\kappa = 0.71$ annotations. Against a best reported system of 0.7362, this leaves a gap of roughly 23.7 points that annotation noise does not explain. The plateau reflects genuine modelling difficulty—or, as we argue next, a metric that stops rewarding progress.

7 What Can Micro-F1 Measure?

At $n = 10{,}200$ and $p \approx .735$, the sampling standard error of a single micro-F1 score is 0.44 points. Because micro-F1 weights every item equally, the maximum change any class can induce is bounded by its share of the test set. Figure 3 plots these bounds.

Sexism accounts for 29 of 10,200 test items. A system with perfect recall on sexist content and a system that never predicts the class differ by at most **0.28 points** of micro-F1—below one standard error, and smaller than the 0.34-point spread separating the entire top five in 1A. Religious Hate (179 items, 1.75 points) is only marginally better placed.

For a hate-speech benchmark this is a design failure, not a statistical curiosity: the evaluation offers no incentive to detect the categories most plausibly tied to real-world harm, and cannot report whether any system does. The majority baseline is 0.5638, so the measurable range above it is 17.2 points, of which Sexism occupies 1.6%. Reporting macro-F1 alongside micro-F1 would cost nothing and resolve this.

8 Label-Schema Coherence

Subtask 1C labels type, severity, and target jointly, permitting internal consistency checks (Table 4).

Condition	n	% test
type=None, severity≠Little to None	0	0.00
type=None, target≠None	0	0.00
type≠None, **target=None**	**342**	**3.35**
type≠None, severity=Little to None	986	9.67

Table 4: Internal consistency of subtask 1C test labels.

The 986 items combining a hate type with Little to None severity are a *tension* rather than an error: casual profanity may reasonably be Profane with little hateful intent. The 342 bearing a hate type but no target are harder to defend, since hostility aimed at nobody contradicts the scheme. We flag these as the priority stratum for re-annotation rather than counting them as confirmed errors.

The dataset is otherwise clean. Across all 50,746 items we find zero duplicate texts carrying conflicting labels, zero verbatim train/test leakage, zero empty or sub-three-word items, and zero items lacking Bangla script. Against the 52.8% and 66.1% error rates Jin et al. (2026) report for English text-to-SQL benchmarks, BanglaMultiHate is a markedly better-constructed resource. Our criticism is of the evaluation, not the data.

9 Recommendations

1. **Report confidence intervals.** One line of code; it would have shown these five-way ties immediately.
2. **Release per-system predictions,** without which McNemar is uncomputable outside the organising committee.
3. **Report macro-F1 alongside micro-F1** on skewed label distributions, so rare classes stay visible.
4. **Validate schema coherence at release.** Table 4 is mechanical.

5. **Declare ties.** Ordering noise manufactures a false narrative of progress.

10 Conclusion

The dataset is well constructed; the evaluation cannot support the rankings it reports. The top five systems are tied in all three subtasks, the plateau is not annotation noise, and the metric is blind to the rarest hate category.

Limitations

Our strongest claims (§4) rest on published scores and the released test set, and hold without further assumptions. Three limitations qualify the rest.

First, we do not have the shared task's per-system predictions. §5 substitutes ten systems of our own, but these are classical models spanning 0.4984–0.6776 micro-F1, below the 0.7362 of the transformer ensembles they stand in for. Discordance between two strong transformers fine-tuned from the same pretrained checkpoint could be lower than anything we measured, since they share an initialisation our systems do not. We regard 3.6% as an empirically grounded floor rather than a proven one; obtaining the organisers' prediction files would settle the question directly, and we invite them to release these. Second, the ceiling estimate in §6 assumes annotators err independently given the true label and adopts a simplified two-parameter error model. Real annotator errors may be correlated in ways neither model captures, which would lower the ceiling; we report the pessimistic correlated-error variant alongside the uniform one, but both remain models. Third, our coherence analysis (§8) identifies items whose labels are mutually inconsistent, which is not the same as identifying which of the labels is wrong. Resolving that requires the manual re-annotation stage this paper does not yet report: a 495-item stratified sample, three blind annotators, and the C0–C4 difficulty taxonomy adapted from Jin et al. (2026). Finally, this audit covers one shared task in one language and does not establish how common these patterns are.

Ethics Statement

This work analyses an existing, publicly released dataset of hate speech under its CC BY-NC-SA 4.0 licence; we redistribute no data, only analysis code. We name teams only as they appear in the published overview paper, and our findings are not criticisms of those teams: the systems may well differ in quality, and our point is precisely that the evaluation cannot tell us. We note that a metric insensitive to sexist content risks under-serving those most affected by it, which motivates §7.

We thank the BLP-2025 organisers, whose decision to release data, gold labels, and scorer is what made this audit possible; we would encourage other shared tasks to match that standard.

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